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Weekly exercises

One exercise per lesson, set on the Friday and due the following Friday. Each lives with its lesson, in `Lessons/NN_topic/Exercises/`.

They are assessed, and they are deliberately small: two to three hours for a student who attended the lesson. The point is regular contact with the material, not a second project.

Standard format

Every exercise states:

- **Goal** — what you will be able to do afterwards
- **Dataset** — named, with the exact loading snippet. Never “use your own data”
- **Tasks** — numbered and unambiguous
- **What to hand in** — a notebook named `{surname}_{NN}.ipynb`
- **Assessment criteria** — with weights

Common assessment criteria

Criterion	Weight
Methodological correctness	40%
Implementation	30%
Interpretation and communication	30%

As with the project, **methodology outweighs results**. An exercise that reaches a modest number by a sound route scores better than one that reaches a good number by an unsound one.

How they build towards the project

The exercises are not independent. Taken in order they walk through the same sequence the project demands, one step at a time:

Lessons	What the exercises establish
1-2	Frame a problem; explore and prepare a dataset
3-4	Fit and evaluate a first model; choose a metric that fits the problem

Lessons	What the exercises establish
5	Design an evaluation before modelling — the pivot of the course
6-7	Compare model families fairly
8-9	Handle unlabelled data; train a network
10	Diagnose a failing model

By Lesson 5 you should be able to write your project’s evaluation plan. By Lesson 7 you should have a draft comparison. The project should not be started in November.

Late work and resubmission

Late submissions are accepted up to the following lesson at reduced credit. An exercise returned with “resubmit” is not a failure — it means a methodological error was found that is worth fixing before it reaches your project, which is exactly what these are for.

Solutions are published after the due date in this folder.